Pedro Nunes Cardoso

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Software Engineer | Full Stack Developer | Systems Architect

Professional Summary

Software Engineer with experience in the architecture and development of data-driven SaaS systems, integration of critical APIs, and construction of automation pipelines. Experience in system design decisions for scalable architectures, relational and vector data modeling, and integration of Artificial Intelligence into real business workflows for decision support and operational automation.

Technical Skills

Backend: Java, Spring Boot, Spring Security, JPA/Hibernate, Node.js, TypeScript, Express
Frontend: React, Vite, Tailwind CSS, Radix UI, Framer Motion
Databases: PostgreSQL, MySQL, MongoDB, SQLite, Supabase, pgvector
Cloud/DevOps: Cloudflare Workers, Vercel, Railway, GitHub CLI, CI/CD
AI: Google GenAI, RAG, embeddings, autonomous agents
Architecture: MVC, monolith vs microservices, Clean Architecture, TDD

Systems Engineering

System Design & Scalable Architectures
Event-driven Systems
API Design & Middleware Engineering
Data Pipeline Orchestration
Fault Tolerance & Idempotency
Observability & Debugging Systems
Distributed Systems Concepts

Professional Experience

Software Engineer | BeHealth (Oct 2025 – Present)

Architecture and development of the BeHealth Intelligence Platform (BIP), a data-driven SaaS for centralizing critical business and patient journey metrics.

- Definition of modular and scalable architecture with domain separation and data flows

- Implementation of reliability patterns such as idempotency, failure handling, and data consistency in critical flows
- Construction of integration pipelines between CRM and external systems via APIs and Webhooks
- Implementation of security layer with JWT, RBAC, sensitive data encryption, and multi-factor authentication (2FA)
- Exposure of operational metrics to support strategic decision-making

Software Engineer & Systems Architect | Indefinidos

- Development of an AI agent with autonomous execution of real actions in internal systems (CRM, emails, document generation, and process automation)
- Creation of a JSON/Regex-based execution engine for intent interpretation and real-time automated action triggering
- Implementation of a vector memory system (RAG) using pgvector for semantic business context retrieval
- Integration of the AI agent directly into operational workflows, acting as an active system automation layer

Software Engineer | Capiwallet (Capstone Project - UFRRJ)

- Development of a digital wallet backend focused on transactional integrity and financial operation security
- Implementation of concurrency control and data consistency in critical operations
- Integration with Pix API (Banco do Brasil)
- Relational database modeling and layered architecture
- Comparative analysis of monolithic architectures for design validation

Education

Bachelor’s Degree in Computer Science – UFRRJ (2026)